Technical Manual — FlowEngine (IDS Dataset Generator)

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> **Versión en español:** ../es/manual_tecnico.md

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> Identifiers, menu text and console messages are kept in Spanish, exactly as they appear in the source code.

1. Introduction

FlowEngine is a Python console tool that **generates network-traffic datasets** for training and evaluating Intrusion Detection Systems (IDS). It:

- Captures live traffic (or reads `.pcap/.pcapng` files) with **Scapy**.
- Groups packets into **bidirectional flows** and produces **91 columns** per flow, **64 of them numeric features** (size and timing statistics, TCP flags, TLS/JA3, DNS, HTTP, entropy, etc.).
- Labels each flow **manually** (the operator types the label live) or **automatically** with a pre-trained **Random Forest** model (`modelo_rf_ids.joblib`).
- Lets you **filter and derive variables** on the resulting dataset.

- Supports **distributed capture**: a *Director* PC discovers agents (sensors) on the LAN, orders them to capture (immediately or scheduled) and receives the resulting CSV files.

1.1 Executables

Executable	Source	Role
`FlowEngine.exe`	`flow_engine.py` (+ modules)	Main menu application. Used interactively on the **Director**; launched in `--captura-remota` mode on **agents**.
`SensorAgente.exe`	`agente_persistente.py`	Window-less resident service that listens for network orders and starts captures. Only used on **agent** machines.

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2. System requirements

2.1 Operating system

OS	Support	Additional requirements
Windows 10/11 (64-bit)	Primary	**Npcap** installed in *WinPcap API-compatible* mode. Nmap optional. Run as Administrator.
Linux (Debian/Ubuntu)	Supported	Run with `sudo` (root). `tcpdump` and `nmap` (auto-installed via `apt-get`).

2.2 Development software

Component	Recommended version	Purpose
Python	3.10 – 3.12	Interpreter
scapy	≥ 2.5	Packet capture and dissection
numpy	≥ 1.24	Flow statistics
pandas	≥ 2.0	Advanced filters and model feature alignment
scikit-learn	**same version used to train the model**	Random Forest prediction
joblib	≥ 1.3	Model loading
cryptography	≥ 41	Required by Scapy's TLS layers
tkinter	bundled with Python	File/folder picker dialogs
PyInstaller	≥ 6.0	Building executables

> ⚠️ The `.joblib` model must be loaded with the **same scikit-learn version** it was trained with; otherwise it may fail or predict incorrectly.

2.3 Automatic dependency installation

When run as a script (`.py`), `instalar_dependencias_python()` checks every library with `importlib.util.find_spec` and installs missing ones with `pip`. In the compiled executable (`sys.frozen`) this step is skipped because libraries are bundled.

`check_and_install_tools()` checks system tools:

- **Windows:** looks for `C:\Windows\System32\Npcap` and `C:\Program Files (x86)\Nmap\nmap.exe`. If missing, runs `npcap-1.88.exe` / `nmap-7.99-setup.exe` located **next to the executable**.
- **Linux:** requires root (`os.geteuid() == 0`) and installs `nmap` and `tcpdump` with `apt-get` if absent.

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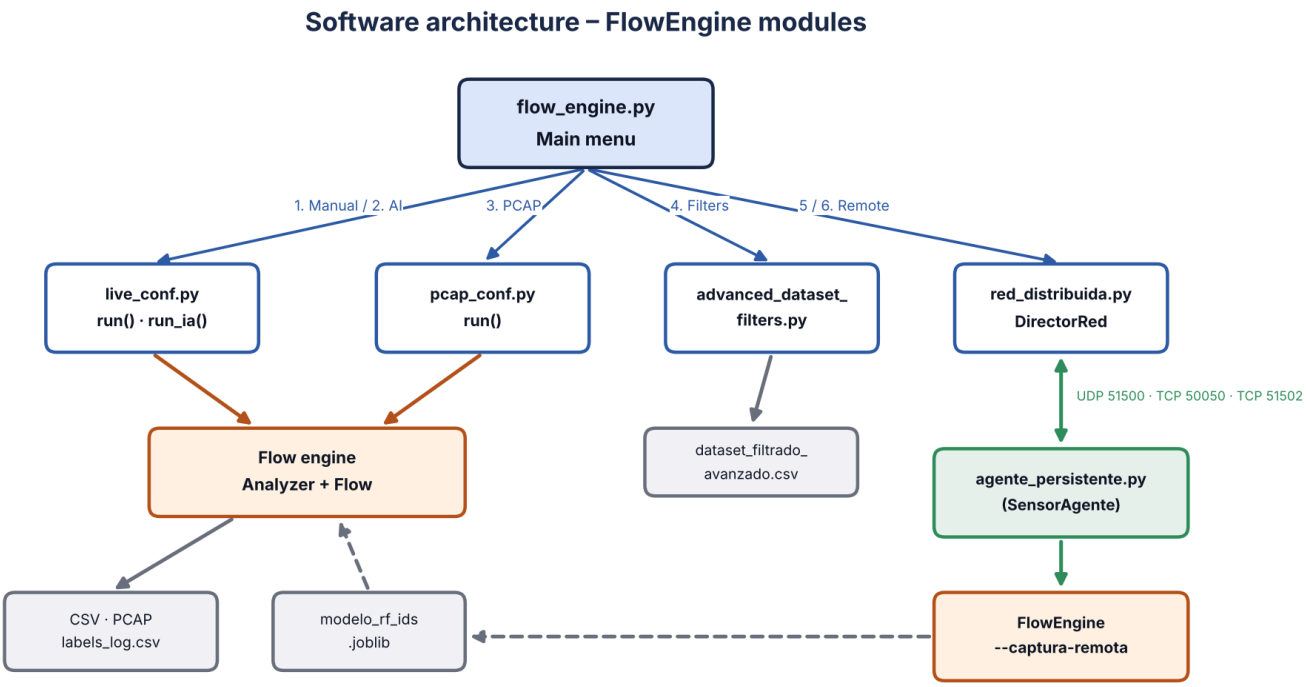
3. Project structure

```
FlowEngine/
├── src/                          # Main program (all .py files together)
│   ├── flow_engine.py            # Entry point: main menu and --captura-remota mode
│   ├── live_conf.py              # Live capture (manual / AI / remote) + flow engine
│   ├── pcap_conf.py              # Offline PCAP processing (same engine)
│   ├── advanced_dataset_filters.py  # Filters and derived variables on a CSV
│   ├── red_distribuida.py        # Director side: discovery, orders, reception
│   ├── agente_persistente.py     # Agent side: UDP/TCP service that starts captures
│   └── compilar.py               # PyInstaller build
├── training/
│   └── entrenar_modelo.py        # Trains modelo_rf_ids.joblib (Random Forest)
├── reproducibility/              # Rebuilds the papers' numbers and figures
├── tools/
│   ├── etiquetar_con_ground_truth.py  # Labels a CSV with the UNSW-NB15 ground truth
│   ├── generar_diagramas.py      # Regenerates docs/img diagrams
│   └── build_pdfs.sh             # Regenerates docs/pdf
├── lab/
│   └── lab_attacks.sh            # Attack-traffic generator for the lab
├── legacy/                       # Earlier versions no longer used
├── datasets/ · models/ · installers/   # README only: data on Zenodo; model on GitHub Releases
└── docs/                         # ES/EN manuals, diagrams and PDFs
```

The `src/` modules must live **in the same folder**: `flow_engine.py` imports them directly (`import live_conf`, ...) and `compilar.py` looks for files in the current directory.

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4. Software architecture



4.1 `flow_engine.py` execution modes

Mode	How it is triggered	What it does
Interactive	`FlowEngine.exe` with no arguments	`check_and_install_tools()` → `preparar_red_distribuida()` (opens TCP 51502 receiver) → `menu_principal()`
Remote capture	`FlowEngine.exe --captura-remota --duracion N --id ID --director-ip IP`	Invoked by the agent. Runs `red_distribuida.ejecutar_captura_y_reportar()` with `live_conf.capturar_remoto` and uploads the CSV to the Director.

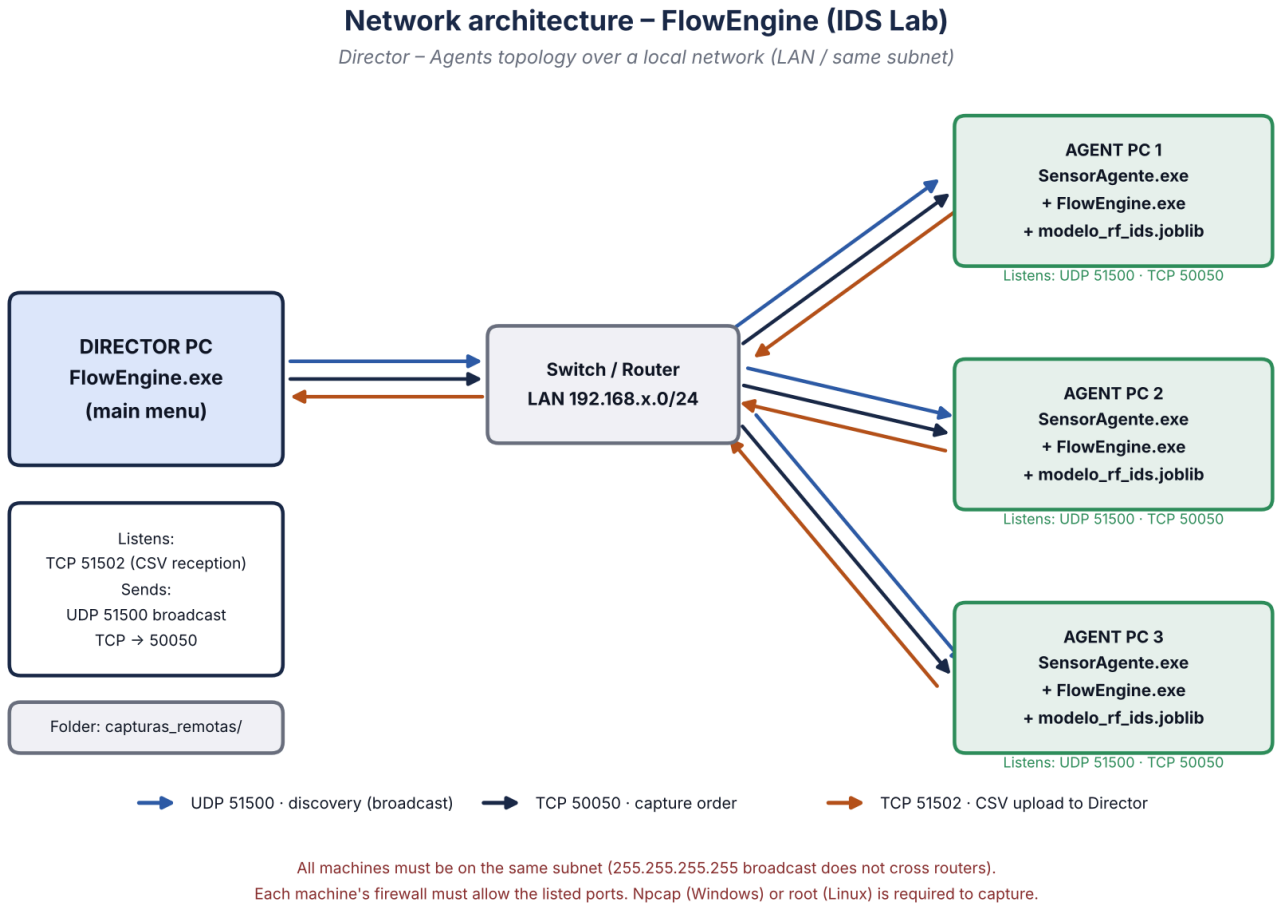
4.2 Concurrency model

Thread	Where	Function
Sniffer (main)	`Analyzer.run()`	`scapy.sniff(prn=Analyzer.process)`
Keyboard	`input_listener()`	Changes current label or stops (`stop` / `salir`)
Reaper	`Analyzer._reaper_loop()`	Every 5 s closes flows idle > `flow_timeout`
Auto-stop	`capturar_remoto.auto_stop()`	Stops a remote capture after `duracion` seconds
TCP receiver	`DirectorRed.iniciar_receptor_reportes()`	Server on 51502; one thread per received file
Agent UDP / TCP	`agente_persistente.py`	Discovery thread + command server; one thread per connection

The shared `Analyzer.flows` dictionary is protected by `threading.Lock()`.

5. Network architecture

5.1 Topology



The system follows a **Director – Agents** model on a **single local network**:

- **Director PC:** runs `FlowEngine.exe` interactively. Discovers agents, sends orders and stores incoming CSV files in `capturas_remotas/`.
- **Agent (sensor) PC:** runs `SensorAgente.exe` in the background. `FlowEngine.exe` and `modelo_rf_ids.joblib` must sit next to it.

5.2 Ports and protocols

Port	Protocol	Listens on	Direction	Purpose
51500	UDP	Agent (`0.0.0.0`)	Director → broadcast `255.255.255.255`	Agent discovery (ping / reply)
50050	TCP	Agent (`0.0.0.0`)	Director → Agent	Capture orders (immediate or scheduled)
51502	TCP	Director (`0.0.0.0`)	Agent → Director	Upload of the resulting CSV

Constants defined in `red_distribuida.py` and `agente_persistente.py`:

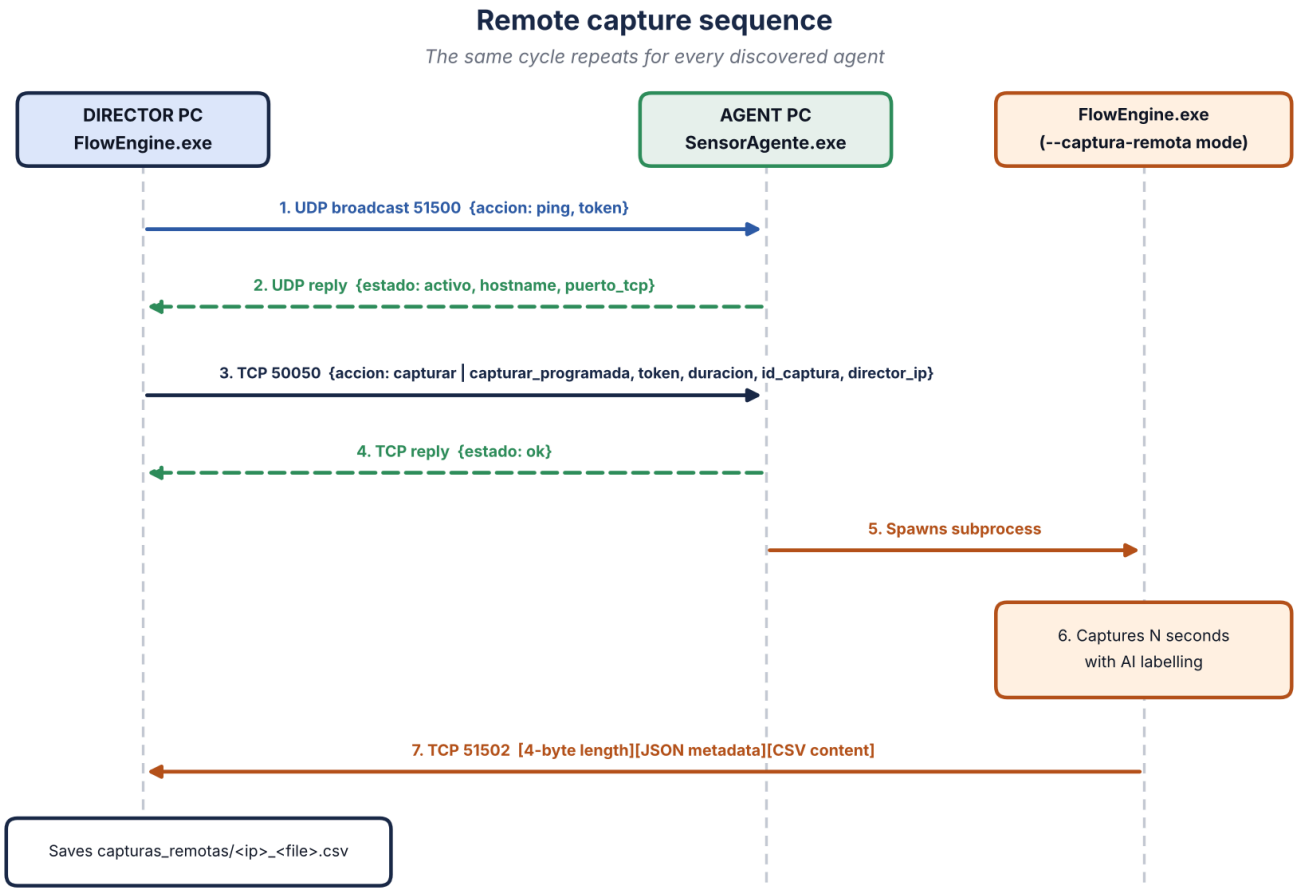
```
PUERTO_UDP_DESCUBRIMIENTO     = 51500
PUERTO_TCP_COMANDOS           = 50050   # agente_persistente.py
PUERTO_TCP_RECEPCION_REPORTES = 51502   # red_distribuida.py
TOKEN_RED                     = _cargar_token_red()
CARPETA_CAPTURAS_REMOTAS      = "capturas_remotas"
```

Network token. `_cargar_token_red()` (present in both modules) looks for it in this order:

1. Environment variable `FLOWENGINE_TOKEN`.
2. File `token_red.txt` next to the program (`.py` or `.exe`), token on the first line.
3. Default `MI_TOKEN_SECRETO_123` (public; testing only).

The Director and **every** agent must use the same token.

5.3 Remote capture sequence



5.4 Message formats

Discovery (Director → broadcast, UDP 51500):

```
{"accion": "ping", "token": "MI_TOKEN_SECRETO_123"}
```

Agent reply (UDP, unicast to Director):

```
{"estado": "activo", "hostname": "PC-LAB-01", "puerto_tcp": 50050}
```

The Director waits **3 s** for replies (`descubrir_agentes(tiempo_espera=3)`).

Capture order (Director → Agent, TCP 50050):

```
{
  "accion": "capturar",            // or "capturar_programada"
  "token": "MI_TOKEN_SECRETO_123",
  "duracion": 60,
  "director_ip": "192.168.1.10",
  "id_captura": "prueba_01",
  "retraso_segundos": 3600,        // only for capturar_programada
  "fecha_ejecucion": "2026-11-03 20:30"
}
```

Result transfer (Agent → Director, TCP 51502):

Bytes	Content
0 – 3	Metadata length *L*, unsigned big-endian integer (`struct.pack('>I', L)`)
4 – 4+L	JSON `{"nombre_archivo": "remoto_<id>_<epoch>.csv", "tamano": <bytes>, "token": "<TOKEN_RED>"}`
rest	Raw CSV content, sent in 4096-byte chunks

The Director rejects the connection if the token does not match, sanitises `nombre_archivo` (no folders or invalid characters) and stores it as `capturas_remotas/<agent_IP>_<nombre_archivo>`. If fewer bytes than announced arrive, it reports an incomplete transfer.

5.5 Network requirements

1. All machines on the **same subnet** (the `255.255.255.255` broadcast does not cross routers).
2. Synchronised clocks (scheduled captures compute the delay from the **Director's** clock).
3. Inbound firewall rules:

Windows (PowerShell as Administrator):

```
# On every AGENT
New-NetFirewallRule -DisplayName "FlowEngine UDP 51500" -Direction Inbound -Protocol UDP -
LocalPort 51500 -Action Allow
New-NetFirewallRule -DisplayName "FlowEngine TCP 50050" -Direction Inbound -Protocol TCP -
LocalPort 50050 -Action Allow
# On the DIRECTOR
New-NetFirewallRule -DisplayName "FlowEngine TCP 51502" -Direction Inbound -Protocol TCP -
LocalPort 51502 -Action Allow
```

Linux (ufw):

```
sudo ufw allow 51500/udp && sudo ufw allow 50050/tcp   # agent
sudo ufw allow 51502/tcp                                # director
```

6. Module reference

6.1 `flow_engine.py` — Entry point

Function	Description
`instalar_dependencias_python()`	pip-installs missing libraries (script mode only).
`check_and_install_tools()`	Checks Npcap/Nmap (Windows) or root + nmap/tcpdump (Linux).
`preparar_red_distribuida()`	Creates `DirectorRed` and starts the TCP 51502 receiver in a daemon thread.
`ejecutar_modo_captura_remota()`	Parses `--captura-remota --duracion --id --director-ip`, runs capture + upload.
`menu_principal()`	Menu loop (options 0–6). Each option runs inside `try/except` so errors never close the program.

6.2 `live_conf.py` — Live capture

Function / class	Description
`run()`	Capture with **manual** labelling. Asks for path, CSV name and PCAP name. Defaults to `<program folder>/Capturas`.
`run_ia()`	Same as `run()` but loads `modelo_rf_ids.joblib` and sets `MODO_ETIQUETADO = "modelo"`. Restores manual mode at the end.
`capturar_remoto(duracion, ruta, id, nombre_pcap)`	Timed, non-interactive capture, always AI-labelled. Returns the CSV path.
`input_listener()`	Keyboard thread: new label → `flush_all_active()` + entry in `LABEL_LOG`.
`get_active_interface()`	Interface of the default route (`conf.route.route("0.0.0.0")`).
`_sanitizar_nombre_archivo()`	Strips paths and invalid characters; enforces the extension.
`_ruta_disponible()`	If the file exists and is locked (Excel, antivirus), uses an alternate name with a time suffix.
`cargar_modelo_ids()` / `_predecir_etiqueta()`	Loads and applies the model (see §9).
`analizar_tls_clienthello()` / `calcular_ja3()`	TLS ClientHello parser and JA3 fingerprint (MD5, GREASE values removed).
`_entropia()`	Shannon entropy of the payload (0–8 bits/byte).
`detectar_protocolo()` / `detectar_service_y_payload()`	Protocol and service classification + DNS/HTTP/TLS metadata.
`Flow`	Accumulates per-flow statistics and builds the CSV row (`compute()`).
`Analyzer`	Manages active flows, CSV/PCAP writing, reaper and label log.

6.3 `pcap_conf.py` — Offline processing

Contains a copy of the engine (`Flow`, `Analyzer`, parsers) adapted to file input. `run()` opens a dialog to pick the `.pcap/.pcapng`, another for the output folder, asks for the CSV name (default `pcap_resultado1.csv`) and streams packets with `PcapReader` (the file is never loaded entirely into memory), printing progress every 10,000 packets.

No output PCAP is written (`save_pcap: False`) and **the resulting CSV has no `Label` column** (there is no labelling in offline mode).

> Note: since the engine exists twice (`live_conf.py` and `pcap_conf.py`), any feature change must be mirrored in both to keep datasets comparable.

6.4 `advanced_dataset_filters.py` — Filters

Function	Description
`seleccionar_dataset()`	Tkinter dialog to pick a CSV.
`convertir_columnas_numericas()`	Converts to numeric only the columns where **every** non-empty value is a number; text columns (`State` , `Service` , `JA3` , `HTTP *` ...) are kept.
`traducir_filtro()`	Translates Wireshark-like syntax into `pandas.query` : `tcp` , `udp` , `&&` → `and` , `||` → `or` , `ip.src` , `ip.dst` . `tcp.port == N` / `udp.port == N` means *protocol and (source or destination port) = N*, as in Wireshark (supports `== != > < >= <=`).
`crear_variable()`	Creates a column with `df.eval(expression)` .
`aplicar_filtro()`	Applies the filter and prints remaining rows.
`guardar_dataset()`	Saves `dataset_filtrado_avanzado.csv` next to the original CSV.

6.5 `red_distribuida.py` — Director side

Element	Description
`obtener_ip_local()`	Finds the outbound interface IP (UDP socket to 8.8.8.8, no data sent).
`DirectorRed.descubrir_agentes()`	UDP broadcast and 3-second reply collection.
`DirectorRed.enviar_orden_captura()`	Sends the JSON order over TCP (10 s timeout).
`DirectorRed.iniciar_receptor_reportes()`	TCP server on 51502 (`listen(10)`), one thread per connection.
`DirectorRed.manejar_recepcion_archivo()`	Reads length + metadata + content, validates the token, sanitises the name and stores the file.
`menu_captura_instantanea()`	Menu option 6.
`menu_programar_captura()`	Menu option 5. Validates `YYYY-MM-DD HH:MM` (24 h) and that the date is in the future.
`ejecutar_captura_y_reportar()`	Agent side: runs the capture and uploads the CSV (30 s timeout).

6.6 `agente_persistente.py` — Agent service

- Sets the working directory to the executable's folder and logs activity to `sensor_agente.log` .
- **UDP thread (51500):** replies to `ping` messages carrying a valid token.
- **TCP server (50050):** validates the token and handles `capturar` (immediate) or `capturar_programada` (thread that waits `retraso_segundos` , then starts the capture).
- `iniciar_motor_captura()` launches `FlowEngine.exe` (Windows) or `FlowEngine` (Linux) with `--captura-remota ...` (or `python flow_engine.py ...` in script mode) with `subprocess.Popen` .
- Built with `--noconsole` : no window; stop it from Task Manager.

6.7 `compilar.py` — Build

Checks/installs PyInstaller, deletes `build/`, `dist/` and old `.spec` files, then builds:

Output	PyInstaller command
`dist/FlowEngine(.exe)`	`--onefile --clean --name FlowEngine flow_engine.py`
`dist/SensorAgente(.exe)`	`--onefile --clean --noconsole --name SensorAgente agente_persistente.py`

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7. Flow engine

7.1 Flow identification

Each packet is assigned to a flow by its **5-tuple** `(source IP, destination IP, source port, destination port, protocol)`. The reverse tuple is also looked up so flows are **bidirectional** (`FWD` = direction of the first packet, `BWD` = reply).

- No IP layer: 802.11 addresses (`addr2/addr1`) or Ethernet MACs are used.
- ICMP: the echo `id` is used as a pseudo-port so separate pings between the same hosts are not merged.
- Detected protocols: `TCP`, `UDP`, `ICMP`, `ARP`, `IP`, `802.11_Beacon/Auth/Deauth/ProbeReq/Other`, `OTHER`.

7.2 Lifecycle (when a row is written)

A flow is closed and written to the CSV on whichever comes first:

1. It reaches `min_packets` (5) packets in total.
2. It is **terminal**: RST seen, or FIN seen in both directions.
3. It exceeds `flow_timeout` (30 s) without traffic (checked by the *reaper* every 5 s).
4. The operator **changes the label** (all active flows are closed with the previous label).
5. The capture **stops** (final flush).

> **Design note:** because a flow closes upon reaching `min_packets`, a long connection produces **several consecutive rows** (~5-packet windows) rather than one row per connection. Raise `min_packets` if you need whole-connection flows.

7.3 TCP state

A state is recorded per packet and `State` takes the **most frequent** one: `S0` (SYN), `S1` (SYN-ACK), `EST` (ACK), `FIN`, `RST`. For UDP/ICMP: `CON` if there was a reply, `INT` otherwise.

7.4 Service detection

Priority: DNS (Scapy layer) → HTTP (Scapy layer) → well-known port (the lower of both ports) → TLS by signature (`0x16`) → `other`.

Mapped ports: 21 ftp, 22 ssh, 23 telnet, 25 smtp, 53 dns, 80 http, 110 pop3, 123 ntp, 137/138 netbios, 139/445 smb, 143 imap, 443 https, 1433 mssql, 1883 mqtt, 1900 ssdp, 3306 mysql, 3389 rdp, 5432 postgresql.

7.5 Scan signatures

`Flag Signature` flags `null_scan` (no flags), `xmas_scan` (FIN+PSH+URG) or `fin_scan` (FIN only); otherwise `none`.

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8. CSV data dictionary

Group	Columns	Description
Identification	`Start Time`, `End Time`, `Source IP`, `Destination IP`, `Source Port`, `Destination Port`, `Protocol`, `State`, `Service`, `SNI`	`HH:MM:SS.ffffff` time, 5-tuple, state, service and TLS SNI
Volume	`Flow Duration`, `Src Bytes`, `Dst Bytes`, `Src Packets`, `Dst Packets`, `Flow Bytes/s`, `Flow Packets/s`, `Src Load`, `Dst Load`, `Bytes Per Pkt`, `Size Diff`, `Byte Ratio`	Duration (s), bytes/packets per direction and rates
TTL	`Src TTL`, `Dst TTL`, `Src TTL Var`, `Dst TTL Var`	TTL mean and variance per direction
Timing	`Sjit`, `Djit`, `Sintpkt`, `Dintpkt`, `Fwd IAT Min`, `Fwd IAT Max`, `Bwd IAT Min`, `Bwd IAT Max`, `Active Mean`, `Idle Mean`	Jitter, inter-arrival time and active/idle periods (`active_gap` = 1 s)
TCP handshake	`tcprtt`, `synack`	SYN→ACK and SYN→SYN-ACK (s)
Context (60 s window)	`ct_srv_src`, `ct_srv_dst`, `ct_dst_ltm`, `ct_src_ltm`	Count of recent flows per source/destination/service (UNSW-NB15 style)
TCP flags	`SYN Count`, `ACK Count`, `FIN Count`, `RST Count`, `PSH Count`, `URG Count`, `ECE Count`, `CWR Count`, `Fwd/Bwd PSH Count`, `Fwd/Bwd URG Count`, `Flag Signature`	Flag counters and scan signature
Sizes	`Pkt Size Mean`, `Pkt Size Var`, `First Pkt Size`, `Fwd/Bwd Pkt Len Min/Max/Mean/Std`	Packet length statistics
Window / retransmission	`Init Fwd Win`, `Init Bwd Win`, `Zero Window Count`, `Fwd Retrans`, `Bwd Retrans`	Initial TCP window, zero windows and retransmissions (same `seq` with payload)
IP	`Frag Count`, `DF Count`	Fragments and *Don't Fragment* packets
Content	`Payload Entropy Mean`	Mean payload entropy (≈7–8 suggests encrypted/compressed)
ICMP	`ICMP Type`, `ICMP Code`	First value seen
DNS	`DNS Qname`, `DNS Qtype`, `DNS Rcode`, `DNS Ancount`, `DNS Qname Len`, `DNS Qname Entropy`	Useful for DGA / DNS tunnelling detection
HTTP	`HTTP Method`, `HTTP Host`, `HTTP URI`, `HTTP User-Agent`, `HTTP Status`, `HTTP Content-Length`	Plain-text HTTP only
TLS	`TLS Version`, `JA3`	ClientHello version and JA3 fingerprint
Class	`Label`	Manual label or model prediction (absent from CSVs produced by `pcap_conf.py`)

Non-applicable fields are left empty (not 0) so they are not mistaken for a real value.

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9. AI model integration

- **File:** `modelo_rf_ids.joblib`, looked up **in the program's own folder** (the path is never asked).
- **Requirement:** the model must expose `feature_names_in_`, which happens when it is trained on a `pandas.DataFrame`.
- **Alignment:** for each flow a one-row DataFrame is built with the `feature_names_in_` columns in that order. Missing or text columns (IPs, service, ...) are filled with **0**.
- **Result:** the prediction (`str`) is written to `Label`. Special values on failure: `sin_pandas`, `sin_prediccion`, `error_prediccion`.

Training: use `training/entrenar_modelo.py`:

```
python training/entrenar_modelo.py --datos datasets --salida training/modelo_salida
```

The script merges every CSV in the folder, normalises labels (`MAPA_ETIQUETAS`), drops non-numeric columns (IPs, service...), tunes hyper-parameters with `RandomizedSearchCV` (15 combinations × 3-fold stratified cross-validation) and saves:

File	Content
`modelo_rf_ids.joblib`	Final model
`columnas_features.json`	List and order of the features used
`matriz_confusion.png`	Confusion matrix on the test set
`importancia_caracteristicas.png` / `.csv`	Feature importance

Because it trains on a DataFrame whose column names match FlowEngine's output, the resulting model works directly with `_predecir_etiqueta()`.

10. Configuration parameters

`CONFIG` dictionary (in `live_conf.py` and `pcap_conf.py`):

Key	live_conf	pcap_conf	Description
`csv_file`	`live_resultado.csv`	`pcap_resultado1.csv`	Default CSV name
`min_packets`	5	5	Packets needed to close a flow
`time_window`	60	60	Window (s) for `ct_*` features
`flow_timeout`	30	30	Idle time (s) before closing a flow
`active_gap`	1.0	1.0	Gap (s) separating active from idle periods
`save_pcap`	`True`	`False`	Store raw traffic in a `.pcap`
`pcap_file`	`captura.pcap`	—	Default PCAP name

11. Generated files

File	Location	Content
`<name>.csv`	Chosen folder (default `Capturas/`)	Flow dataset
`<name>.pcap`	Same	Raw traffic (written with `sync=True`, survives abrupt exits)
`labels_log.csv`	Same	`timestamp_unix, timestamp_legible, label` for every label change
`filas_no_guardadas.csv`	Same	Rows that could not be written after 5 retries (locked CSV)
`dataset_filtrado_avanzado.csv`	Folder of the filtered CSV	Output of option 4
`capturas_remotas/<ip>_remoto_<id>_<epoch>.csv`	Director	CSV received from each agent
`capturas_remotas/remoto_<id>_<epoch>.pcap`	Agent	Local PCAP of the remote capture (not uploaded)
`sensor_agente.log`	Agent	Service log

12. Build and deployment

12.1 Build

```
cd src
pip install -r ../requirements.txt
python compilar.py
```

Output in `src/dist/`: `FlowEngine(.exe)` and `SensorAgente(.exe)`. On Linux: `chmod +x dist/*`.

> PyInstaller does not cross-compile: build on Windows for `.exe` files and on Linux for Linux binaries.

12.2 Distribution bundle

Director PC:

```
FlowEngine.exe
modelo_rf_ids.joblib      (optional, only for option 2)
npcap-1.88.exe            (installer, downloaded from npcap.com)
nmap-7.99-setup.exe       (optional)
token_red.txt             (recommended: same token on every machine)
```

Agent PC:

```
SensorAgente.exe
FlowEngine.exe            (required: the agent launches it)
modelo_rf_ids.joblib      (required: remote capture always uses AI)
npcap-1.88.exe
token_red.txt             (recommended: same token as the Director)
```

12.3 Agent auto-start (optional)

- **Windows:** create a Task Scheduler task with "Run with highest privileges" and an "At log on" trigger, or put a shortcut in `shell:startup` (the user must be an administrator to capture).
- **Linux (systemd):**

```ini
# /etc/systemd/system/sensoragente.service
[Unit]
Description=FlowEngine SensorAgente
After=network-online.target
[Service]
WorkingDirectory=/opt/flowengine
ExecStart=/opt/flowengine/SensorAgente
Restart=on-failure
User=root
[Install]
WantedBy=multi-user.target
```

---

13. Security

Risk	Detail	Recommendation
Public default token	If not configured, `MI_TOKEN_SECRETO_123` is used, visible on GitHub	✅ Now read from `FLOWENGINE_TOKEN` or `token_red.txt` (excluded by `.gitignore`). Set it on every machine
Unencrypted traffic	Orders, token and CSV files travel in plain text	Use only on isolated lab networks or add TLS (`ssl` module)
Unauthenticated receiver	Port 51502 used to accept files from any host	✅ Fixed: metadata carries the token and the Director validates it
Unsanitised file name	`nombre_archivo` comes from the client	✅ Fixed: `_nombre_seguro()` strips folders and invalid characters
Elevated privileges	Capturing requires admin/root	Run only on lab machines
Sensitive data	PCAP/CSV files may contain real IPs, domains, URIs and user agents	Never commit real captures (see `.gitignore`)

> **Ethical use:** only capture traffic on networks you own or are explicitly authorised to monitor.

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14. Known issues and recommended improvements

14.1 Fixed in version 1.1

#	Issue	Location	Effect it had	Fix applied
1	Inconsistent delay key: the Director sent `retraso_segundos`, the agent read `delay_segundos`	`agente_persistente.manejar_conexion`	Every scheduled capture started after 60 s, not at the chosen time	The agent reads `retraso_segundos` (still accepts `delay_segundos`)
2	Binary name hard-coded as `FlowEngine.exe`	`agente_persistente.iniciar_motor_captura`	On compiled Linux the agent could not find the engine	`FlowEngine.exe` on Windows, `FlowEngine` on Linux
3	`tcp.port` / `udp.port` mistranslated (`Protocol == 'TCP'.port == 80`)	`advanced_dataset_filters.traducir_filtro`	Syntax error when filtering by port	Translated before `tcp` / `udp`; compares source or destination port
4	`convertir_columnas_numericas` emptied text columns	`advanced_dataset_filters`	`Start/End Time`, `State`, `Service`, `Flag Signature`, `DNS Qname`, `HTTP *` became empty in the filtered CSV	Only fully numeric columns are converted
5	Examples used the non-existent `Packet Length Mean` column	`advanced_dataset_filters`	Confused users	Examples use `Pkt Size Mean` / `Fwd Pkt Len Max`
6	Hard-coded token, unauthenticated receiver, unsanitised file name	`red_distribuida`, `agente_persistente`	See §13	Configurable token validated on reception; sanitised name; exact metadata reads
7	Hard-coded `D:\Descargas\...` paths	`training/entrenar_modelo.py`, `tools/etiquetar_con_ground_truth.py`	Only worked on the original PC	`--datos/--salida` and `--csv/--gt/--salida` arguments

14.2 Pending (recommended improvements)

#	Issue	Location	Effect	Suggestion
8	Duplicated engine	`live_conf.py` and `pcap_conf.py`	Risk of datasets diverging	Move `Flow` / `Analyzer` into a shared module
9	Remote PCAP is not uploaded	`ejecutar_captura_y_reportar`	Only the CSV reaches the Director	Also upload the `.pcap` if needed
10	`Start Time` / `End Time` without date	`Flow.compute`	Ambiguous for captures spanning midnight	Include date or epoch
11	`pcap_conf.py` CSVs have no `Label`	`pcap_conf.Flow.compute`	Labelling must happen afterwards	Fixed-label or model option
12	Unencrypted communication	`red_distribuida`, `agente_persistente`	Token and data readable on the wire	TLS via the `ssl` module

15. Troubleshooting

Symptom	Likely cause	Fix
`ERROR CRÍTICO DE RED: SCAPY NO DETECTA NPCAP`	Npcap just installed or without WinPcap mode	Restart the program; if it persists, reinstall Npcap ticking *WinPcap API-compatible Mode*
`No se detectaron agentes en la red`	Agent not running, firewall, or different subnet	Check `SensorAgente.exe` in Task Manager, open UDP 51500, same subnet
`Fallo de conexión con <ip>`	TCP 50050 blocked	Firewall rule on the agent
Director never receives the CSV	TCP 51502 blocked or Director IP changed	Firewall rule on the Director; check `sensor_agente.log`
`Archivo rechazado de <ip>: token inválido`	Agent and Director use different tokens	Same `token_red.txt` (or `FLOWENGINE_TOKEN`) on every machine
`No se encontró el modelo IA`	`modelo_rf_ids.joblib` missing next to the executable	Copy it to the program folder (and the agent's)
`Label = error_prediccion`	Different scikit-learn version or incompatible features	Retrain using FlowEngine's column names
`'file' está en uso`	CSV open in Excel	Close it; the program automatically switches to an alternate name
Linux: `necesitas permisos de root`	Run without sudo	`sudo ./FlowEngine`

16. Companion tools

Tool	Purpose
`training/entrenar_modelo.py`	Trains `modelo_rf_ids.joblib` (§9).

Tool	Purpose
`tools/etiquetar_con_ground_truth.py`	Labels a CSV produced from a UNSW-NB15 PCAP by joining it with `NUSW-NB15_GT.csv` on the 4-tuple (IPs and ports). Adds `attack_cat` and `label` (0/1). Usage: `python tools/etiquetar_con_ground_truth.py --csv flows.csv --gt NUSW-NB15_GT.csv --salida labelled.csv` (needs `tqdm`).
`lab/lab_attacks.sh`	Generates the lab's attack traffic (nmap, hydra, hping3, slowloris, ffuf, sqlmap, bettercap) and pauses before each attack so the operator can switch the label in FlowEngine. **Isolated, self-owned lab only.** See lab/README.md.
`reproducibility/`	Five scripts that, from the Zenodo PCAP and CSV files, rebuild the label audit, the corrected dataset, the Random Forest results (cross-validation and unseen captures), the CICFlowMeter comparison, latency and the papers' figures.
`tools/generar_diagramas.py` · `tools/build_pdfs.sh`	Regenerate this documentation's diagrams and PDFs.